

Student Perceptions of Large Language Models Use in Self-Reflection and Design Critique in Architecture Studio

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Abstract. This study investigates the integration of Large Language Models (LLMs) into the feedback mechanisms of the architectural design studio, shifting the focus from generative production to reflective pedagogy. Employing a mixed-methods approach with architecture students at the Singapore University of Technology and Design, the research analyzes student perceptions across three distinct feedback domains: self-reflection, peer critique, and professor-led reviews. The findings reveal that students engage with LLMs not as authoritative instructors, but as collaborative "cognitive mirrors" that scaffold critical thinking. In self-directed learning, LLMs help structure thoughts and overcome the "blank page" problem, though they are limited by a lack of contextual nuance. In peer critiques, the technology serves as a neutral mediator, mitigating social anxiety and the "fear of offending". Furthermore, in high-stakes professor-led juries, students utilize LLMs primarily as post-critique synthesis engines to manage cognitive overload and translate abstract academic discourse into actionable design iterations.

Keywords: Architectural Education, Design Studio Pedagogy, Large Language Models, Generative AI in Education, Design Critique.

1 Background and motivation

The architectural design studio is widely regarded as the discipline's signature pedagogy, a model historically rooted in the École des Beaux-Arts tradition and centered on learning through iterative production and public critique [1]. Within this pedagogical paradigm, Donald Schön's [2] concept of the reflective practitioner has been foundational, positioning design learning as a "reflective conversation with the situation." This reflective dialogue is typically enacted through three interconnected feedback channels: self-reflection, peer critique, and professor-led jury reviews. Together, these mechanisms are intended to support the development of critical judgment, design reasoning, and professional identity.

Despite their centrality, the effectiveness of these traditional feedback mechanisms has been increasingly questioned due to persistent psycho-social and cognitive barriers. The design jury, in particular, operates as a high-stakes, public performance that frequently induces anxiety rather than productive reflection. Instead of enabling

reflection-in-action, students often experience juries as asymmetrical power encounters, prompting defensive postures and performative compliance rather than openness to critique [3]. These dynamics undermine the pedagogical intent of the studio and contribute to broader well-being concerns within architectural education.

Peer critique, while theoretically positioned as a low-stakes and collaborative alternative, is similarly constrained by social friction. Hyland and Hyland [4] describe a pervasive “fear of offending,” whereby students prioritize social harmony over critical honesty. This phenomenon commonly referred to as the Mum Effect results in feedback that is overly positive, vague, or non-actionable, significantly diminishing its instructional value. Empirical studies further suggest that peer feedback is often distorted by friendship bias, where students hesitate to critique peers with whom they have close social ties or only value feedback from those perceived as artistically superior [5].

These socio-emotional barriers are compounded by cognitive constraints inherent to architectural design. Design tasks are widely characterized as wicked problems, imposing a high intrinsic cognitive load due to their open-ended, ill-defined nature [6]. When students must simultaneously process large volumes of verbal feedback, manage contradictory critiques, and navigate social stressors, extraneous cognitive load increases substantially. Prior work has shown that such overload inhibits students’ ability to synthesize, retain, and operationalize feedback, ultimately limiting deep learning [7].

In response to these long-standing challenges, architectural educators have experimented with a range of non-AI pedagogical interventions aimed at restructuring the social dynamics of critique. Frameworks such as the Critical Friends protocol redefine the role of the critic as a trusted interlocutor rather than an evaluator, formalizing feedback into “warm,” “cool,” and “hard” categories to reduce interpersonal friction [8]. Other strategies including anonymous peer review, silent critiques, gallery walks, and flipped classroom models seek to lower social risk, democratize participation, and shift critique from performative confrontation to reflective engagement [9,10,11]. While these approaches demonstrate measurable benefits, they remain labor-intensive, context-dependent, and difficult to scale consistently across studio settings.

Concurrently, the rapid integration of Generative Artificial Intelligence (GenAI) and Large Language Models (LLMs) into architectural education has opened new avenues for pedagogical innovation. However, the dominant body of research has focused almost exclusively on LLMs as tools for generative production and technical optimization rather than as supports for pedagogical practices such as facilitation or reflective learning [12]. Studies have emphasized their role in early stage ideation, rapid visual brainstorming, and overcoming creative inertia [13], as well as in convergent tasks such as structural code generation, compliance checking, and urban design automation [14].

Only recently has scholarship begun to explore LLMs as feedback-generating tools in educational contexts, with emerging evidence suggesting their potential to reduce perceived power imbalances, particularly in doctoral supervision and academic writing [15]. Nevertheless, a critical research gap persists in architectural education. There is limited empirical understanding of how LLMs might operate within the social and cognitive ecology of the design studio not merely as generators of content, but as

mediators of critique and scaffolds for reflection. Specifically, it remains unclear how students perceive LLMs as neutral interlocutors capable of bypassing the fear of offending in peer critique, or as cognitive aids capable of synthesizing fragmented and sometimes contradictory jury feedback.

Research in Human–Computer Interaction (HCI) provides promising but underexplored parallels. Studies on AI-mediated collaboration indicate that LLMs can function as psychological buffers, reducing social anxiety and encouraging participation by offering a non-judgmental interaction partner [16]. In conflict resolution and online moderation, LLMs have demonstrated a capacity to maintain perceived impartiality and de-escalate emotionally charged exchanges, often outperforming human moderators in perceived neutrality [17]. Additionally, work on machine-generated feedback suggests that receiving critique from an AI prior to human evaluation can lower ego defensiveness and increase receptivity to subsequent feedback [18]. Despite these insights, their application to the architectural design studio particularly in mitigating critique-related anxiety and cognitive overload remains largely speculative.

At the same time, critical scholarship cautions against unreflective AI adoption. Empirical studies on creativity and LLMs use indicate a risk of generative homogenization, where reliance on AI raises baseline performance but reduces collective diversity, leading to convergent design outcomes [19]. This tension underscores the need to conceptualize LLMs not as authoritative critics or replacements for human judgment, but as reflective partners whose role is carefully constrained and pedagogically aligned.

Against this backdrop, the present study investigates the role of LLMs as mediators within architectural design feedback processes. Rather than focusing on AI-driven production, the study centers on students’ lived experiences of critique and reflection, and on how AI-mediated systems might restructure feedback interactions without displacing human agency. Specifically, the study addresses the following research questions:

1. How do architecture students currently engage with traditional feedback loops of self-reflection, peer critique, and professor-led reviews?
2. What are the perceived strengths and weaknesses of integrating LLMs into these distinct feedback domains?
3. What pedagogical roles and system features do students envision for LLMs to effectively support learning while preserving critical judgment?

By addressing these questions, this research responds to urgent concerns surrounding student well-being, feedback literacy, and cognitive overload in architectural education. It contributes a reframing of LLMs as partners in reflection rather than tools of production, proposing a hybrid feedback model in which AI supports low-level synthesis and emotional buffering while human educators retain responsibility for high-level conceptual and ethical mentoring. In doing so, the study advances a pedagogically grounded framework for the ethical and reflective integration of AI in the architectural design studio.

2 Methods

2.1 Research Design

This study employed a mixed-methods research design with a sequential explanatory logic of integration to investigate architecture students' reflections, perceived challenges, strengths, and expectations related to critique sessions when using Large Language Models (LLMs). The mixed-methods approach was selected to capture both the breadth of students' experiences across different critique contexts and the depth of their lived experiences within the architecture studio.

The quantitative phase consisted of three online surveys, each focusing on a distinct critique modality commonly encountered in architectural education: (1) self-reflection, (2) peer-to-peer critique, and (3) professor-led critiques. Across the three surveys, data were collected primarily through closed-ended questions, including Likert-scale items and multiple-choice questions, complemented by a limited number of open-ended questions that allowed participants to elaborate on specific responses. The surveys were administered online using Microsoft Forms and could be completed asynchronously at a time convenient for participants. All survey responses were securely stored within the principal investigator's institutional Microsoft account.

Following the survey phase, semi-structured interviews were conducted to further explore the same thematic domains, reflection practices, challenges, strengths, and expectations but in a more open and conversational manner. The interviews aimed to elicit richer accounts of students' experiences in the architecture studio and to gain insight into how future tools leveraging LLMs could better support critique processes. Interviews were audio-recorded with participants' consent and securely stored on the principal investigator's computer.

Consistent with a sequential explanatory design, survey data collection preceded the interviews, which were conducted several days later. Insights from the survey responses informed the focus of the interview discussions, allowing qualitative data to contextualize and expand upon patterns observed in the quantitative findings.

After data collection, quantitative survey data were analyzed using descriptive statistical techniques to identify trends across critique modalities, while qualitative data from both open-ended survey responses and interview transcripts were analyzed using a thematic analysis approach. The thematic analysis was done by analyzing the interview script and encoding the sentences or phrases into key phrases. A list of themes emerged after the key phrases were mapped based on their frequency of the code mentioned, enabling the quantitative data to be supported by the participants' views. This combined analysis enabled the identification of recurring themes related to students' reflections, challenges, strengths, and expectations concerning critique practices and the use of LLMs.

The overarching goal of this mixed-methods design was to derive key design requirements and features for future tools whether standalone LLMs-based systems or LLMs-integrated platforms that could meaningfully enhance self-reflection, peer critique, and professor-led critique within the architectural design studio.

2.2 Participants and sampling

A total of 22 students enrolled at the Singapore University of technology and Design in the Architecture and Sustainable Design Pillar participated in the study, with an average age of 23.5 years. As illustrated in **Fig 1**, the gender distribution was nearly balanced, comprising 50.0% Female and 45.5% Male, with the remaining 4.5% preferring not to disclose their gender. Regarding the current level of architectural study, **Fig 2** indicates that the largest group of participants consisted of Year 3 students (45.5%), followed by Year 1 (22.7%) and Year 2 (13.6%), while Master's level students and those in other categories each accounted for 9.1%.

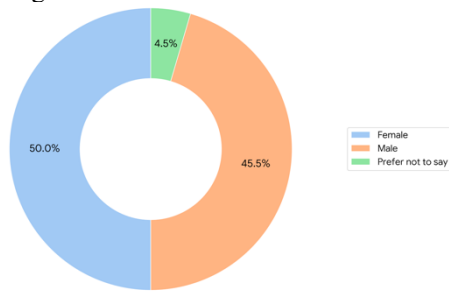


Fig. 1. Distribution of Gender

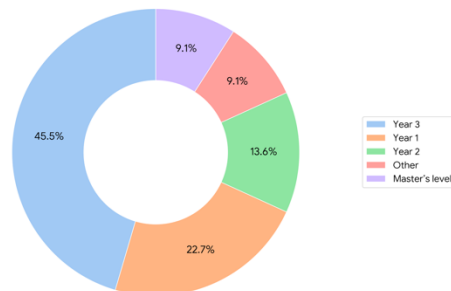


Fig. 2. Current Level of Architectural Study

Participation was subject to two inclusion criteria. First, participants were required to have prior experience with architectural studio critiques, including self-reflection, peer-to-peer critiques, or professor-led critiques. Second, participants were required to have used Large Language Models (LLMs) for any aspect of their architectural studio work, including reflection, critique preparation, or design-related tasks. These criteria ensured that respondents could meaningfully reflect on both critique practices and the role of LLMs within studio contexts.

Participants were recruited through institutional email announcements and university messaging application groups. Participation was entirely voluntary, and no academic or evaluative consequences were associated with participation or non-participation. Students who met the inclusion criteria and consented to participate were able to complete the surveys and, if selected, take part in the follow-up semi-structured interviews.

2.3 Data collection

The study employed a mixed-method data collection strategy designed to capture both the breadth of student experiences and the depth of their pedagogical engagement. The process was divided into two distinct phases: a quantitative survey series followed by qualitative semi-structured interviews.

Survey Instruments: data collection commenced with the administration of three distinct surveys, designed to assess different pedagogical dimensions:

- Self-Learning ([Appendix, Table 1](#))
- Peer- Learning ([Appendix, Table 2](#))
- Professor-Led Learning ([Appendix, Table 3](#))

Each survey instrument was structured around four core thematic sections: Reflections, Challenges, Strengths, and Expectations. The questions were predominantly closed-ended to establish a quantitative baseline of student sentiment, supplemented by selected open-ended questions to capture initial qualitative insights.

Semi-Structured Interviews. Following the completion of the surveys, participants were subjected to a semi-structured interview protocol ([Appendix, Table 4](#)). These interviews were designed to deepen the inquiry into the four themes established in the surveys. Some of the questions were reused in the interview to probe spontaneous response that the online survey could not.

2.4 Procedure

The study followed a sequential explanatory procedure, with quantitative data collection preceding qualitative data collection. Data collection began with the administration of three online surveys, each corresponding to a different critique modality: self-reflection, peer-to-peer critique, and professor-led critiques. The surveys were distributed simultaneously and administered via Microsoft Forms. Participants completed the surveys on their personal computers at a time of their convenience, with each survey requiring approximately 15 minutes to complete. Throughout the surveys, participants were encouraged to provide additional detail in open-ended questions to further elaborate on their experiences and perspectives.

Prior to accessing the surveys, participants were presented with an informed consent form detailing the objectives of the study, the voluntary nature of participation, potential risks, data handling procedures, and confidentiality measures. Only participants who provided consent were able to proceed with the surveys. Following the survey phase, a subset of participants was invited to participate in follow-up semi-structured interviews. Interview participants were selected from among survey respondents who met the inclusion criteria and expressed willingness to take part in an interview. The interviews were conducted several days after survey completion to support the sequential explanatory design.

Semi-structured interviews were conducted online via Zoom and lasted approximately 30–40 minutes on average. Participants were encouraged to expand upon their survey responses, provide concrete examples from their studio experiences, and elaborate on their reflections, challenges, strengths, and expectations regarding critique sessions and the use of Large Language Models. With participants' consent, interviews were audio recorded and subsequently transcribed for analysis. All recordings and transcripts were securely stored on the principal investigator's computer.

3 Results

3.1 Self-learning

Reflections: Usage Patterns and The Role of AI

The study reveals a ubiquitous adoption of Large Language Models (LLMs) among architecture students, with 100% of respondents in the self-learning survey confirming they have used tools like ChatGPT in their design studies. However, the data suggests a distinct shift in how these tools are perceived not as authoritative instructors, but as collaborative aids. The vast majority of students, representing 58% of the sample group, perceive the LLM's primary role as a "Discussion partner," compared to only 16% viewing it as a "Tutor" and 6% as a "Critic" (See Fig 3). This aligns with interview findings where the respondent described using LLMs to develop concepts and aligning the design goals with the brief. Reflection is treated as an iterative process rather than a final checklist. Students utilize LLMs most frequently "during design iteration" (41%) and "after critiques" (37%), rather than solely before final submission (See Fig 4). In the context of high-stakes professor-led critiques, the reliance shifts toward synthesis. 95% of respondents identified the ideal role of AI as a "Post-critique reflection partner". This suggests students are using LLMs to deconstruct and process the dense feedback received during juries, rather than using AI to replace the professor's judgment.

Participant Responses:

"During my design process, I often write descriptions of my project. This helps me gain clarity of thought in my design. I have fed the design brief and my own descriptions into LLMs so I can reflect on whether I have hit the brief, or if I am starting to go off the brief."

"I will usually ask [ChatGPT] for conceptualisation, putting my thoughts into a comprehensive design concept or suggestions to continue from what has been discussed in my consultation with my tutors."

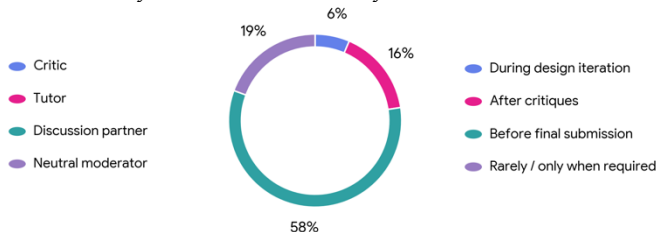


Fig. 3. Perceived Role of the LLMs

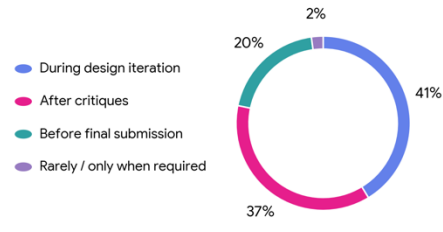


Fig. 4. Timing of LLMs Usage in the Design Process

Challenges: The Specificity Gap and Human Uncertainty

While adoption is high, students face significant challenges regarding the depth of AI feedback and their own confidence in evaluating architectural work. The most prevalent technical limitation is the lack of contextual nuance. 32% of respondents reported that LLMs responses are "too generic," and 36% noted a "misunderstanding of design intent" (See Fig 5). The interview respondents reinforced this, stating that feedback can be too general and superficial to fully capture the specifics of the project. The data reveals that the primary barrier to self-reflection is structural rather than emotional, with

"Uncertainty about what to evaluate" (30%) being the most significant challenge. In contrast, "Emotional discomfort" is negligible at only 5%, suggesting students are willing to critique their work but simply lack the necessary framework to do so. This finding validates the potential utility of LLMs, which can directly address the other major hurdles by providing the specific vocabulary (20%) and objectivity (21%) that students are currently missing (See Fig 6).

Participant Responses:

"One challenge is that LLMs feedback can be too general... There is also a risk of relying on it too much, so I always try to still largely use my own thinking and judgement."

"...LLMs can only understand as much as I do the input and lack subjectivity born from the human relationship and interaction. Therefore, sometimes when I look for the references, LLMs would still give me information that is completely contradictory to what I have been looking for. ..."

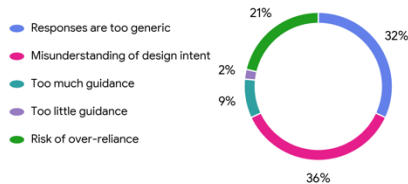


Fig. 5. Reported Limitations of LLMs

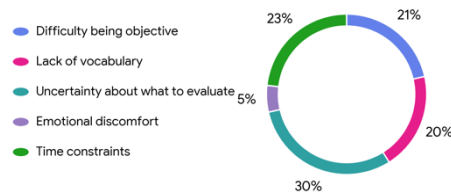


Fig. 6. Top Barriers in Reflection

Strengths: Objectivity, Organization, and Social Safety

The results indicated that the Large Language Models served primarily as a cognitive scaffolding tool rather than a replacement for deep design expertise. Participants overwhelmingly reported that the tool aided in the organization of their ideas, with 97% agreeing or strongly agreeing that it helped structure their thoughts. Furthermore, 92% of respondents noted that LLMs prompted them to consider new questions, effectively preventing "tunnel vision" during the reflection process (see Fig 7). These findings addressed the previously identified barrier of methodological uncertainty, suggesting that LLMs effectively mitigated the "blank page" problem by providing a necessary framework for self-evaluation. When compared directly to human feedback, LLMs demonstrated distinct advantages in communication and emotional safety, though it lagged in practical utility. A significant majority (85%) of participants rated the clarity of the LLM's feedback as superior to that of human tutors, reinforcing its role in supplying the descriptive vocabulary that students often lacked. Additionally, 65% of respondents preferred the emotional impact of the AI, likely due to its perceived objectivity, which 55% rated as better than human critique. However, the tool was viewed as less practically helpful; "usefulness" received the lowest positive rating at 45%, with a notable portion of students finding it only comparable to or worse than human advice. This suggested that while LLMs were an effective, low-stakes partner for initiating reflection and reducing anxiety, it did not yet match the specific, high-level design guidance provided by human mentors.

Participant Responses::

"They are especially useful for giving general objective feedback, breaking down complex peer/professor feedback, and suggesting areas to focus on"

“LLMs are helpful at giving clarity to the project brief and crafting narratives. Oftentimes thoughts are all over the place so LLMs help to reorganise and give your design a clear narrative/clarity (to) others who read it.”

LLMs help me to:

Compared to human feedback, LLM feedback is:

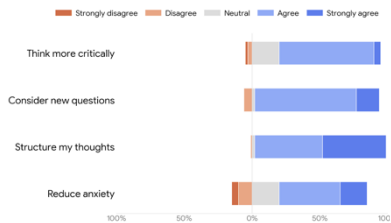


Fig. 7. LLMs perceived strengths in reflection



Fig. 8. LLMs vs. Human Feedback

Expectations: The Demand for Context-Aware Tools

Participants expressed a strong preference for a collaborative approach to AI-mediated reflection. As illustrated in **Fig 9**, an overwhelming 86% of respondents indicated that ideal LLMs should offer "a balance of both" guidance and freedom, rejecting the extremes of total autonomy (9%) or rigid instruction (5%). This finding suggested that students valued the tool as a flexible partner capable of adapting to their workflow rather than a prescriptive tutor.

This demand for balanced support was operationalized in the specific feature requests detailed in **Fig 10**. The highest priorities were "Bias or objectivity checks" (15 students), followed by "Structured reflection frameworks" and "Studio-specific vocabulary" (13 students each). These preferences aligned with the previously identified barriers of subjectivity and lack of terminology, confirming that students sought LLMs that acted as a "cognitive mirror" one that provided structure and neutrality without imposing a specific design direction.

Participant Responses:

“It could serve as a moderator only if it is fed with the relevant information required for the context and has internalised the right materials for greater context to give a quick reference if any questions arise or direct the discussion in the right track to prevent going off track.”

“... a well-defined moderator for discussion with the following features: transcription; tracks all feedback, who made the feedback and who the feedback is addressed to; can summarise feedback from each distinct individual for each project; can draw trends and therefore key ideas.”

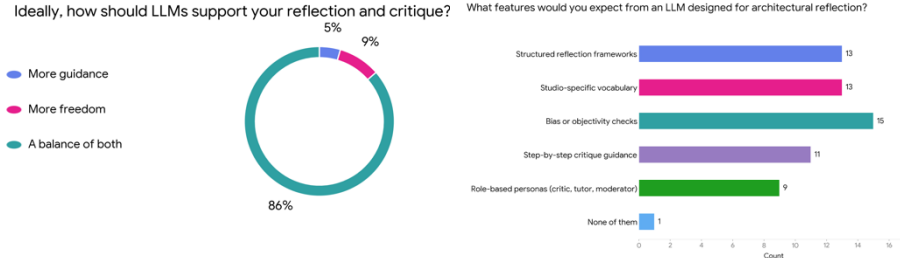


Fig. 9. Preferred Integration of LLMs in Reflection

Fig. 10. Most Desired LLMs Features

3.2 Peer learning

Reflections: Utilization and the Social Context of Critique

The survey results highlight a distinct contrast between the perceived value of peer feedback and the actual adoption of AI tools to support it. While 70% of respondents consider peer critiques "Important" or "Extremely Important" for their learning, the use of Large Language Models (LLMs) to *prepare* for these interactions remains surprisingly low. Only 10% of students reported using LLMs "Yes" (regularly) to prepare peer critiques, with another 15% using them "Sometimes." The vast majority (75%) do not use AI for preparation, which contrasts sharply with the ubiquitous use of AI for self-reflection found in other studio contexts (see Fig 11). Despite this low usage in preparation, the formats of critique suggest an environment ripe for intervention. Peer critiques are predominantly "Informal discussions" (85%) or "Pin-ups" (65%) (see Fig 12). In these unstructured settings, students perceive the LLM's potential role not as a replacement for peers, but as "Knowledge support" (60%) or a "Neutral mediator" (45%). This suggests that while students are not yet habitually using AI to prep for peer reviews, they recognize its potential to stabilize the often fluid and informal nature of student-to-student feedback.

Participant Responses:

"I also use them to better understand, organize, and respond to feedback from my peers..."

"... Perhaps when conversing with peers, we would use chatGPT to open up more conversational thoughts or as a basis to start a discussion of our projects but we would still have our own natural thinking process to work on the design rather than to rely on ChatGPT solely."

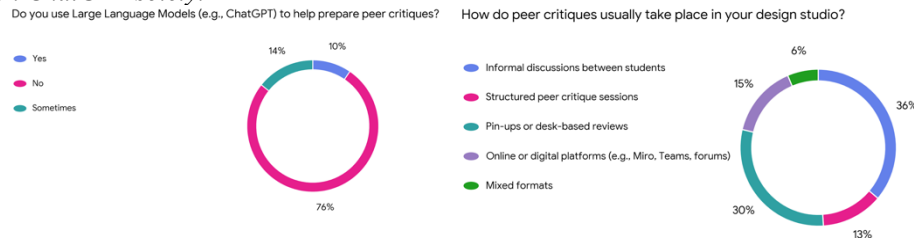


Fig. 11. Frequency of LLMs Usage for Peer Critique Preparation

Fig. 12. Common Formats of Peer Critique

Challenges: Social Anxiety and the "Shared Ignorance"

The primary barriers in peer critique are deeply rooted in social dynamics and confidence gaps. A striking 27% of respondents identified "Lack of confidence in my expertise" as a major challenge when giving feedback, while 23% struggled with "Finding the right architectural vocabulary" (see **Fig 13**) This "shared ignorance," as one respondent phrased it, creates a feedback loop where students hesitate to critique because they feel unqualified or fear offending their peers (20%). On the receiving end, students struggle to filter the noise. 24% reported "Conflicting opinions" as a challenge, and 22% cited "Difficulty deciding what to act on." (see **Fig 14**) When LLMs are introduced to this equation, they bring their own set of challenges. The most significant limitation reported was "Over-generalized feedback" (90%) and a "Lack of contextual understanding" (80%). This indicates that while students struggle with the *social* delivery of feedback, AI currently struggles with the *contextual* specificity required to make that feedback relevant.

Participant Responses: *"Students are generally not as experienced in giving feedback, so it is understandable that feedback might not be the most constructive/objective. We also tend to not give feedback unless asked as there is a fear of 'offending' others which is a common reason. But likewise there is a 'fear of getting negative feedback', likewise people do not ask others for advice as well."*

What do you find most challenging when giving feedback to peers?

- Finding the right architectural vocabulary
- Being objective
- Avoiding emotional or personal judgments
- Lack of confidence in my expertise
- Fear of offending peers

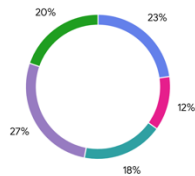


Fig. 13. Human Barriers to Giving Peer Feedback

What challenges do you face when receiving feedback from peers?

- Conflicting opinions
- Feedback that is too vague
- Feedback that feels subjective or biased
- Emotional discomfort or defensiveness
- Difficulty deciding what to act on

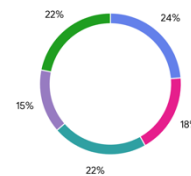


Fig. 14. Challenges in Receiving Feedback

Strengths: Articulation and Emotional Safety

Despite the low usage for preparation, students identify powerful potential benefits for AI in the peer review space, particularly in overcoming the vocabulary and confidence barriers mentioned above. An overwhelming 28% of respondents believe LLMs would be effective at "Helping articulate criticism clearly." This directly addresses students who struggle to find the right words. Furthermore, 23% noted that AI can "Offer multiple perspectives," and 20% believe it can "Support less confident students (see **Fig 15**)." **Fig 16** indicates that students perceive LLMs as a tool for democratization, capable of leveling the playing field in design studios. A clear majority agreed that its integration would result in "More equal participation" and "Reduced dominance by confident students." Most significantly, the data shows a strong consensus of approximately 80% that the tool will foster a "Greater focus on design rather than personality" suggesting that the AI's objectivity effectively decouples the architectural evaluation from the student's social performance.

Participant Responses:

"They may help students feel more prepared and confident when discussing their work."

"I feel that it may become a distraction and should be used after the critiques/feedback from peers and professors only to check if there are other areas of discussion that could be fruitful to the discussion and no repetition of what has been originally thought of critically at that junction."

What do LLMs will do particularly well in peer critique contexts?

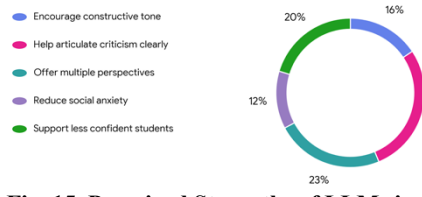


Fig. 15. Perceived Strengths of LLMs in Peer Contexts

The use of LLMs in peer critiques will lead to:

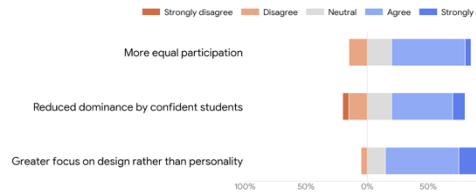


Fig. 16. Usage of LLMs in Peer Critique

Expectations: Scaffolding and Optional Integration

Students envision a future where AI serves as a specialized scaffold rather than a mandatory moderator. There is a strong consensus on the need for domain-specific tools, with 75% of respondents requesting "Studio-specific vocabulary" and 65% asking for "Bias or subjectivity detection" (see Fig 17). The desire for bias detection highlights the students' awareness of the subjective nature of peer feedback and their desire for a tool that can parse helpful critique from personal preference. In terms of implementation, students prefer autonomy. 52% viewed the ideal integration of LLMs as an "Optional individual support tool," whereas only 22% supported the idea of a "Real-time moderator during critiques" (see Fig 18). This suggests that students want to use AI to bolster their own capabilities, checking their bias, refining their vocabulary, and structuring their thoughts before and after engaging with their peers, rather than having an AI intervene in the conversation itself.

Participant Responses:

"Ideally, I see LLMs... as support tools that help organize thinking and make feedback clearer. The main critique... should still come from people."

"It should be a role to reinforce on existing architectural reflection and critique as a checking point only after original and critical thoughts has been made by relevant designers."

What features would you want in an LLM designed specifically for peer critique?

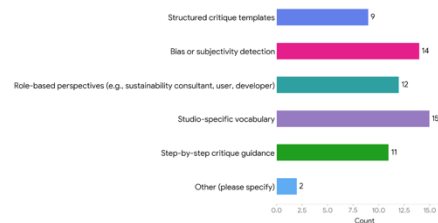


Fig. 17. Most Desired LLMs Features for Peer Critique

Ideally, how should LLMs be integrated into peer critique sessions?

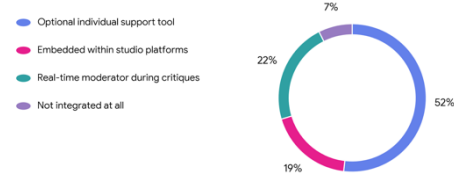


Fig. 18. Preferred Mode of LLMs Integration in peer critique

3.3 Professor led learning

Reflections: The Emergence of the AI Reflection Partner

The survey indicates a significant integration of LLMs into the high-stakes environment of professor-led critiques, with 75% of students reporting that they use tools like ChatGPT to help interpret or reflect on feedback from professors. Unlike peer contexts where usage is lower, the pressure of the jury review drives students to seek support both before and after the event. 70% of respondents use LLMs to prepare for these sessions, primarily to "refine design narratives" (32%) (see Fig 19). However, the most dominant role for AI is retroactive; 95% of students identified the ideal role for LLMs as a "Post-critique reflection partner", helping them process the dense and often overwhelming information received during reviews. This usage pattern correlates with how students naturally manage critique data. 31% of students reflect by "reviewing comments later on", and 25% revisit recordings or written notes (see Fig 20).

Participant Responses:

"I would treat it as feedback from any other peer, where it could possibly help me but I would have to consider how suitable it is for my design. ...I would treat my professor's feedback with more weight given their experience."

"...LLM feedback can feel too generalized at times, so I mainly use it as a starting point for reflection rather than as a final direction."

How do you use LLMs before juries?

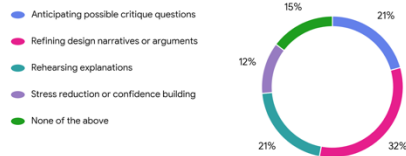


Fig. 19. LLMs Usage in Jury Contexts

After a jury or professor-led critique, how do you usually reflect on the feedback?



Fig. 20. Ideal Role of LLMs in Juries

Challenges: Cognitive Overload and Interpretative Ambiguity

The primary driver for AI adoption in this context appears to be the cognitive difficulty of processing real-time expert feedback. During the jury itself, 24% of students find it challenging to process "Conflicting opinions from different jurors", and 23% struggle with "Time pressure to respond". The complexity of the feedback often leads to confusion, with 23% of respondents noting difficulties with "References to unfamiliar precedents" (see Fig 21). These challenges persist long after the review ends. When trying to make sense of the feedback later, 23% of students struggle with "Ambiguous or abstract comments", and an equal 22% report a "Lack of clear next steps" (see Fig 22). The emotional and cognitive toll is significant, often leading to what one respondent described as a struggle to simply remember what was said while simultaneously trying to defend the work.

Participant Responses:

"One challenge is that LLM feedback can be too general and might not fully capture the specifics of my project."

During juries, what do you find most difficult to process in real time?

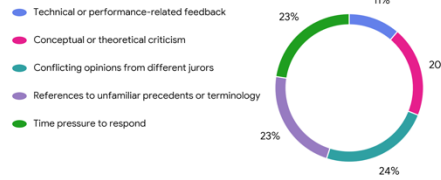


Fig. 21. Real-Time Challenges During Juries

What do you find most challenging when making sense of jury feedback afterward?



Fig. 22. Post-Jury Processing Difficulties

Strengths: Synthesizing Complexity and Bridging Gaps

Students overwhelmingly value LLMs for its ability to impose order on chaos. In **Fig 23**, the highest-rated capability is "Help organize complex feedback". This expectation directly translates into practice: "Summarizing feedback" is the top reported use case in **Fig 24**. This suggests that the primary utility of the AI is not necessarily generating new ideas, but synthesizing the fragmented, unstructured data stream of a verbal jury into a coherent narrative. The data indicates a strong reliance on AI to decode expert language. "Bridge gaps in theoretical or technical knowledge" was the second-highest expected benefit in **Fig 23**. This is mirrored in **Fig 24**, where "Clarifying terminology or references" and "Translating abstract critique into concrete actions" are dominant activities. Students are essentially using LLMs as an on-demand tutor to demystify high-level architectural discourse and convert abstract academic feedback into actionable design steps.

Participant Responses: *"They are especially useful for giving general objective feedback, breaking down complex peer/professor feedback, and suggesting areas to focus on. This helps me prioritize my revision..".*

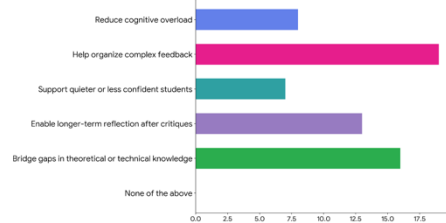


Fig. 23. Perceived Strengths of LLMs in Jury Learning

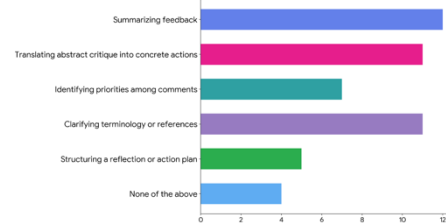


Fig. 24. Specific Post-Jury LLMs Tasks

Expectations: Structure, Alignment, and Actionability

According to **Fig 25**, half of the participants (50%) view the ideal role of LLMs as a "Post-critique reflection partner," significantly outweighing its use as a preparatory tool (30%) or a real-time moderator (20%). Notably, zero participants selected "No role," indicating universal acceptance of AI integration in some form. This desire for retrospective support is driven by a need to organize chaotic feedback. **Fig 26** shows that the most requested features are a "Structured critique breakdown" (17 counts) and "Action-oriented summaries" (14 counts). This suggests students value the AI's ability

to translate unstructured verbal commentary into clear, actionable steps for design iteration.

Participant Responses:

“I would want LLMs to play a sort of mediator role, connecting the bridge between the abstract conceptual advice from the instructors and personal reflection.”

Ideally, what role should LLMs play in professor-led critiques or juries?

- Pre-critique preparation tool
- Post-critique reflection partner
- Real-time moderator (if allowed)
- No role in jury contexts

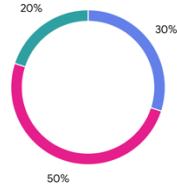


Fig. 25. LLMs preferred role in Professor-led critique

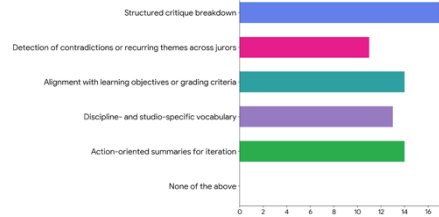


Fig. 26. Desired Features for Jury Support LLMs

3.4 Semistructured interviews

Participants describe their engagement with LLMs across several themes, categorized by their application in self-learning, peer-learning, and professor-led learning contexts. First, the analysis reflects on how students and professionals currently incorporate these tools to navigate design briefs and consolidate scattered thoughts. Second, the analysis identifies the strengths of LLMs, particularly their ability to provide objective technical feedback and rapid information synthesis. Third, it addresses critical weaknesses, such as the lack of spatial intelligence and the risk of stifling original thought. Finally, the section outlines the future expectations of participants, who envision LLMs as unbiased "background assistants" or "sports commentators" that enhance rather than replace the human-centric critique process.

The reflections on using LLMs reveal they are primarily used as a "launchpad" for individual research and conceptualization. In self-learning, participants use these tools to summarize design briefs, ensuring they remain "on the right track" and do not "drift off content". For peer-learning, LLMs are used to structure critiques and open up more "conversational thoughts" during group discussions. When reflecting on professor-led learning, students use LLMs to consolidate and re-organize notes from desk reviews, translating abstract conceptual advice into actionable to-do lists. The "mediator" role in peer helps bridge the gap between the subjective tips from instructors and a student's own personal reflection

The strengths of LLMs across these modalities lie in their speed, objectivity, and capacity for synthesis. For self-learning, the tools offer instantaneous visualization of concepts and the ability to process thousands of pages of technical data in hours which is a task impossible for humans alone. In the context of peer-learning, LLMs can identify bias in reviews, potentially reducing the emotional stress of "poor remarks". Regarding professor-led feedback, LLMs are praised for being "more objective" and consistent, providing a stable "baseline" compared to human opinions that may "change from week to week". Furthermore, for students with auditory processing disorders or dyslexia, the ability to convert vocal feedback into visual text and diagrams is considered "extremely useful".

Conversely, participants also highlight challenges regarding visual intelligence and intellectual depth. For self-learning, LLMs exhibit a "very limited ability" to manipulate or understand geometrical data, often failing to provide value when analyzing architectural screenshots or "cryptic" sketches. In peer and professor-led contexts, the feedback is often criticized for being "too general," "predictable," or "sycophantic," lacking the "drive for novelty" and deep "studio context" that a human expert provides. There is also a pervasive fear that over-reliance will lead to a loss of the "originality thought process," as students might use AI to answer "hard questions" during critiques instead of thinking critically on the spot.

Participants' expectations for the future role of LLMs focus on specialised, integrated assistance rather than creative autonomy. They desire features like speech recognition to automatically transcribe live critiques and tools that can visually track trends in feedback through charts or diagrams. In peer interactions, LLMs are expected to act as moderators that ensure discussions stay "on the right track" and identify "conflicting comments". For professional-level learning, the expectation is for a "background AI tool" integrated directly into CAD environments, providing a "running commentary" similar to modern programming assistants.

While LLMs are powerful tools for information management and technical "checking points," the participants think that they currently lack the spatial reasoning and subjective nuance essential to architectural mastery. One of the most productive roles according to a participant is for AI to be a "sports commentator": a tool that helps beginners make sense of the "game" of design without actively "guiding the student's hand" or replacing the player's own critical judgment. The consensus points to AI being a support to the "mundane data assembly," but the core design decisions and the "uniqueness" of the process must remain firmly with the human designer.

4 Discussion

4.1 Cognitive skills

At the foundational level of *remembering*, LLMs acts primarily as a post-event archival tool that mitigates the cognitive overload of critique sessions. Students frequently reported an inability to retain the sheer volume of information presented during reviews, with one respondent noting the difficulty of "remembering everything" and "catching references I never heard of". In this capacity, AI helps by serving as an external memory bank, converting ephemeral verbal feedback into permanent, reviewable records. Students envision future tools that can "record the feedback clearly" or provide "meeting minute summaries", effectively offloading the burden of memorization so they can focus on the discussion itself. However, LLMs currently fail to assist in the immediate moment of remembering due to processing latency. It does not yet help students "process other ongoing conversations and information in realtime". Until AI tools can listen and transcribe instantly without manual intervention, they remain retrospective archives rather than real-time cognitive aids, leaving students to manage the immediate stress of the critique unaided.

Moving to the level of *understanding*, LLMs serve as a powerful bridge between the novice student and the expert juror. Architectural education is often laden with

complex terminology that can alienate students. Respondents described jurors using "flowery language" or obscure references. LLMs helps significantly here by "explaining terms used" and "bridging the gap between technical expertise and beginner knowledge" This democratizes access to high-level theory, allowing students to grasp concepts that would otherwise be lost. Yet, this understanding is often superficial because LLMs lack deep, studio-specific context. It operates as a dictionary rather than a cultural interpreter, often failing to grasp the "essence of studio concepts unless the prompt is really specific". Consequently, while LLM can define a term, it often misses the specific pedagogical intent or the "why" behind a professor's comment, leading to a disconnect between the definition and its application in the studio.

In the *application* phase, students successfully employ LLMs to structure their workflows and articulate their design narratives. The data indicates that LLMs are rarely used to generate the design itself but is invaluable for "structuring thoughts and discussions into a safe base or skeleton outline". It aids students in "better articulating and translating thoughts", effectively allowing them to rehearse their arguments before high-stakes presentations. This scaffolding gives students a "safe base" to test their explanations. However, LLMs hinder application when it is relied upon too heavily, often yielding "generic responses". Students found that while LLMs can organize an argument, it cannot apply design changes to the project itself; it can talk about architecture but cannot "do" it. Over-reliance here leads to a phenomenon where the verbal presentation improves while the actual design work remains static or disconnected from the narrative.

At the level of *analysis*, LLMs provide a unique benefit by stripping away social and emotional noise. Students value it for "filtering out harsh critiques" and offering "pure objectivity". This allows students to analyze feedback dispassionately, separating their personal feelings from the critique of their work, which is particularly difficult in the heat of a public review. Conversely, LLMs fail to provide the deep, intuitive analysis that comes from human experience. It cannot replace "genuine human feedback with years of unique human experience" because it cannot visually or spatially analyze the design. This limitation leads to a "shared ignorance" where the LLMs might hallucinate a critique that does not align with the visual evidence, proving that while it is an excellent text analyst, it remains a poor design critic.

When *evaluating* their own work, students use LLMs to create a psychological safety net. It serves as a non-judgmental "side buddy" that validates ideas and reduces the anxiety that often paralyzes decision-making. This "space and freedom to have conversations without being judged" is crucial for building confidence. Despite this, LLMs are not accepted as a valid final evaluator of architectural quality. Students explicitly state that "feedback from professors is still more important" and do not trust LLMs to make the final call on merit. While it helps students evaluate their options, it is viewed as inferior to the "human element" of judgment, which is seen as indispensable for assessing the true value of a design.

Finally, at the level of *creation*, LLMs help by acting as a boundless library that acts as a catalyst for ideation. Students envision it aiding creativity by "identifying real-life project references" and suggesting "branches of ideas generation", which is particularly useful when they are stuck. However, this comes with a significant caveat: the risk of standardization. There is a palpable fear among students that using LLMs will lead to "repetitive" outcomes and "less creativity" because the feedback is based on average

data patterns rather than unique, individualistic thought. While AI helps generate a higher volume of ideas, it does not necessarily support the creation of better or more original ones, potentially leading to a homogenization of student work.

4.2 Key design features for future LLMs

Regarding Self-Reflection, students identified a critical disconnect where they design visually but are forced to reflect verbally. Future LLMs must bridge this gap through Multimodal Fluency. To address this, students requested image-to-text analysis tools capable of visual hierarchy analysis, essentially allowing them to input images and receive image-based outputs. A key feature would be an AI that acts as an objective "mirror," scanning a student's drawing to answer whether the visual rhetoric actually conveys their stated intent, enabling them to verify their internal narrative. Furthermore, instead of simply providing answers, the AI should employ Socratic prompting engines. This "Socratic mode" would challenge the student's logic with "what if" scenarios and simulate a desk critique to deepen critical thinking without offering direct solutions.

In the context of Peer Critique, the process is often hindered by social anxiety and a lack of vocabulary, meaning future tools should focus on Social Scaffolding and Bias Mitigation. Students expressed a strong desire for real-time bias detection, effectively a "tone check" that flags overly aggressive or non-constructive language before a peer sends it. Conversely, this tool could help shy students articulate criticism clearly to ensure their valid points are heard. To further overcome the fear of offending peers, collaborative anonymity was suggested. This feature would allow peers to input feedback anonymously, which the AI would then synthesize into a single "studio consensus" report, stripping away personality politics to focus purely on emerging design patterns.

Finally, for Professor-Led Critique, which acts as a high-information and high-stress event, the primary design needs are Cognitive Offloading and Contextual Linking. Students explicitly requested live transcription with speaker diarization, an AI that can listen in real time, recognize who is speaking, and separate a professor's feedback from a peer's comment. This would automatically generate "meeting minutes," freeing students from note-taking so they can maintain engagement and eye contact. Additionally, to address the pain point of professors citing unknown architects, an active referencing system is needed. This "Active Linker" would listen for proper nouns and immediately push visual references to the student's device, instantly filling knowledge gaps. Unlike current chat models that reset sessions, students also want longitudinal progress tracking with "long-term memory." This system would remember feedback from prior reviews and check if the current design has addressed those specific points, acting as a continuity guard against forgotten advice.

4.3 Limitations

While the research provided quantitative and qualitative insights on students' take on the use of LLMs in architecture design studio, the scope of this research is limited by the access to sample size and researcher's time. This study only included the architecture of the students from SUTD for the survey and the semi-structure interview. However, there are only 7 students available for the qualitative interview due to limited budget and time where this study could not compensate the participants for their time.

Hence, the qualitative portion is not statistically significant. Future research is recommended to re-perform this study with more participants and architecture students from other universities.

Furthermore, since all of the participants had the experience in using LLMs in their studio work, there is the potential bias where their attitude toward LLMs is more positive. This bias however, is necessary due to the questions in the survey and interview required the students to draw upon their previous experience in a design studio so that they can provide their insights of their use of LLMs, the perceived strengths and challenges in the context of an architectural design studio. Future research can standardize the nature of the questions to enquire about their experience when using LLMs instead of their attitudes of LLMs.

5 Conclusions

This study confirms that while the adoption of LLMs in the architectural design studio is now ubiquitous, students have fundamentally redefined the tool's pedagogical role from an authoritative instructor to a collaborative partner. Rather than viewing the AI as a tutor or critic, most students perceive it as a discussion partner or cognitive mirror intended to scaffold their own thinking processes. This shift is driven by a preference for a balance between guidance and freedom, where students utilize the technology to articulate vague intuitions and synthesize high-density feedback without surrendering their creative agency or judgment.

Despite high adoption rates, the effectiveness of these tools is currently hindered by significant structural barriers across all feedback domains. In self-directed learning, a "specificity gap" remains the primary obstacle, as generic AI responses often fail to grasp the nuanced design intent of a project. In peer critique settings, "shared ignorance" and social anxiety limit engagement, creating a demand for tools that can help students articulate criticism professionally and filter out personal bias. Furthermore, the cognitive overload inherent in professor-led juries drives students to use LLMs primarily as post-critique synthesis engines, tasked with organizing conflicting opinions and translating abstract academic discourse into actionable next steps.

To address these specific pedagogical gaps, future architectural AI systems must move beyond general text generation to become context-aware, studio-specific support tools. Development should focus on features such as "design intent profiles" that anchor feedback in specific project constraints, thereby solving the issue of generic responses. Additionally, systems should incorporate "Socratic vocabulary engines" and "bias checkers" to help students articulate concepts with greater precision and objectivity, bridging the vocabulary gap identified in both self and peer reflection. Finally, to support the critical post-jury phase, tools must be capable of ingesting and synthesizing unstructured transcripts into prioritized action plans, transforming the chaotic data of a design review into a clear roadmap for iteration.

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APPENDIX

1. Survey and Interview Questions: [click to view](#)
2. Self-learning survey Responses: [click to view](#)
3. Peer Survey Responses: [click to view](#)
4. Professor-led Survey Responses: [click to view](#)
5. Interview Scripts : [click to view](#)

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